

電腦視覺
Computer Vision: from Recognition to Geometry

簡韶逸 Shao-Yi Chien

Department of Electrical Engineering

National Taiwan University

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Computer Vision

- Describe the world that the computer see in one or more images and to reconstruct its properties, such as shape, illumination, and color distribution
- Is it hard? An inverse problem

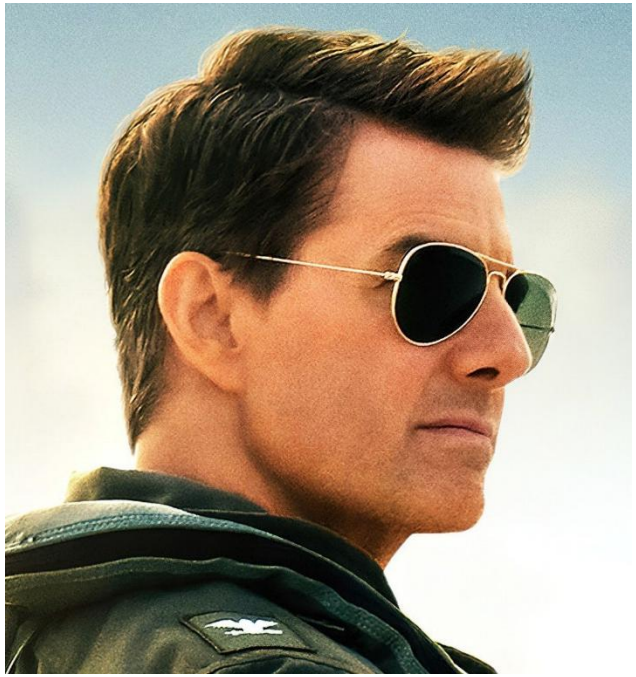


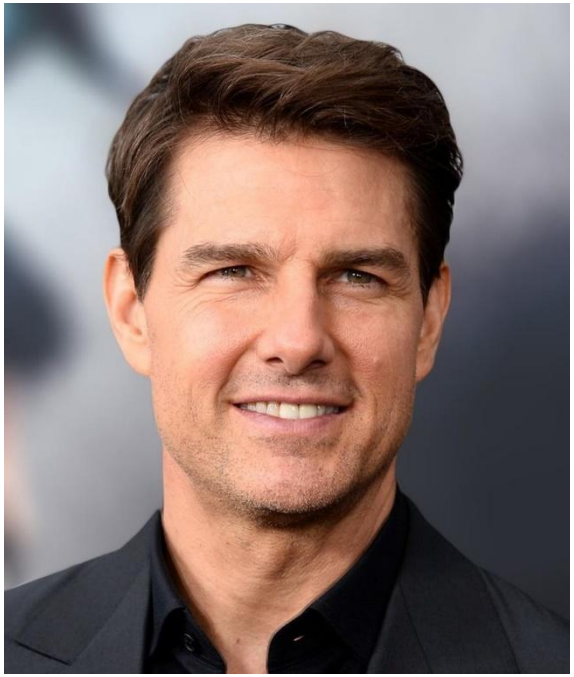
Computer Vision



[R. C. James]

Computer Vision

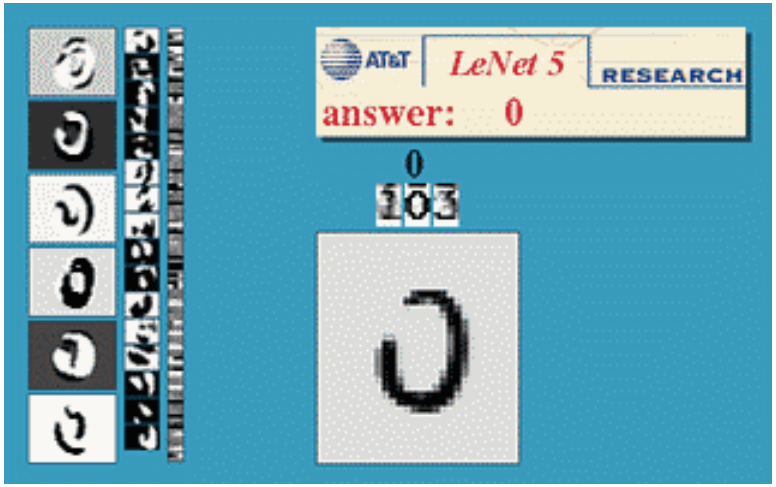




Same Person?

Wide Applications of Computer Vision

- Optical character recognition (OCR)



Digit recognition, AT&T labs
http://www.research.att.com/~yann/



License plate readers
http://en.wikipedia.org/wiki/Automatic_number_plate_recognition

Wide Applications of Computer Vision

- Face detection: in all digital cameras and smart phones



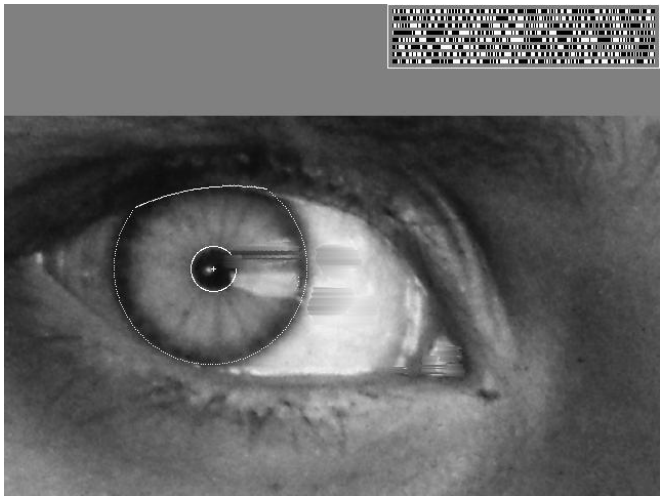
Wide Applications of Computer Vision

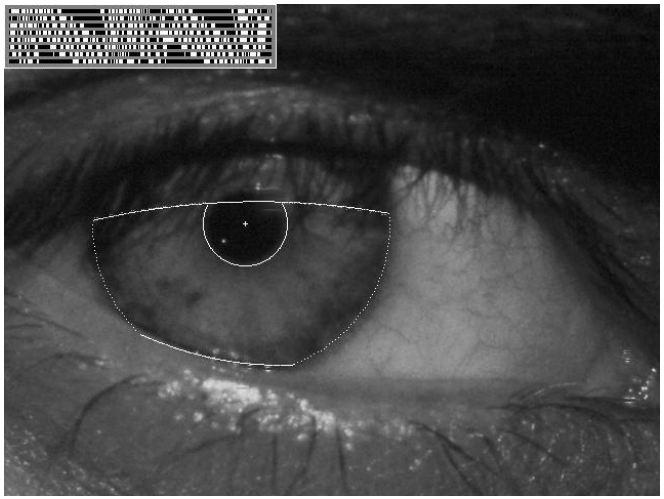
- Iris recognition

  (Vision-based biometrics)



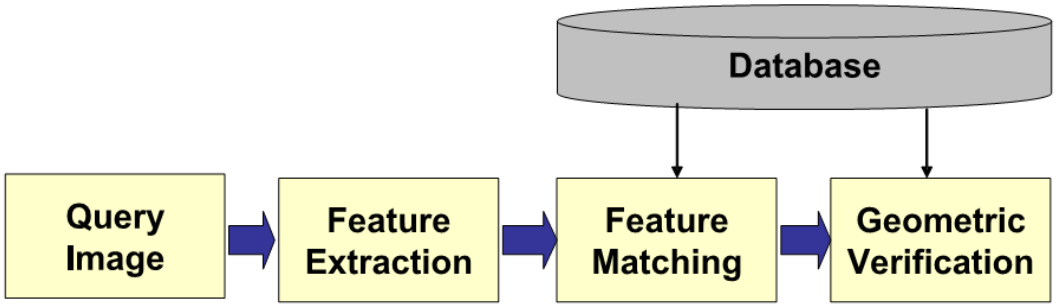
"*How the Afghan Girl was Identified by Her Iris Patterns*" Read the story





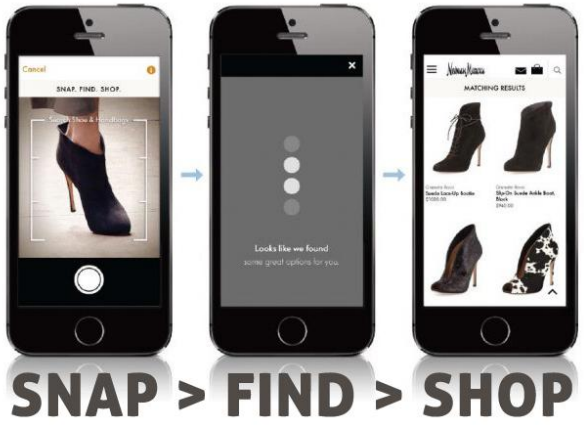
Wide Applications of Computer Vision

- Object recognition





[Girod et al. 2011]



[slyce.it]

Wide Applications of Computer Vision

- Motion capture



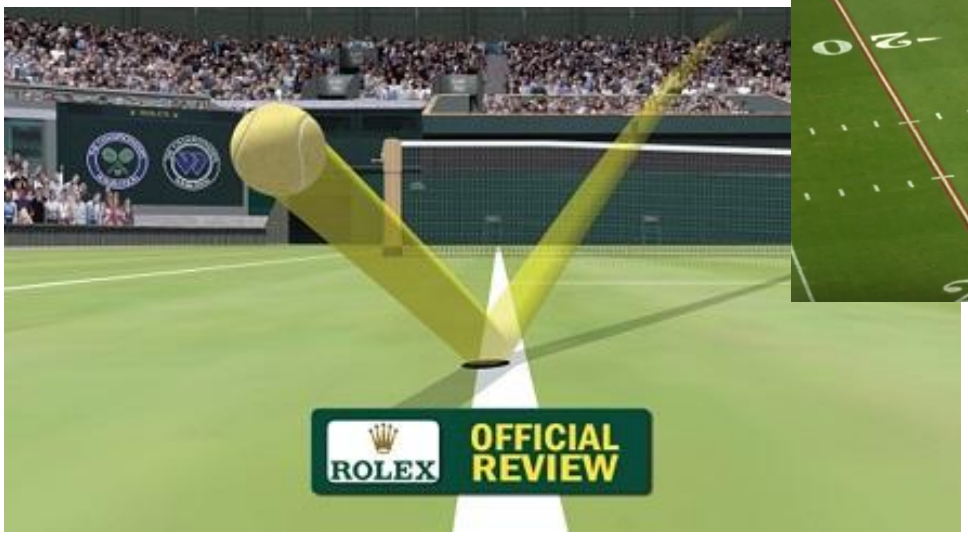
Pirates of the Carribean,
Industrial Light and Magic

Wide Applications of Computer Vision

- Computer vision in sports



Intel: freeD technology



Hawk-Eye: helping/improving referee decisions

Wide Applications of Computer Vision

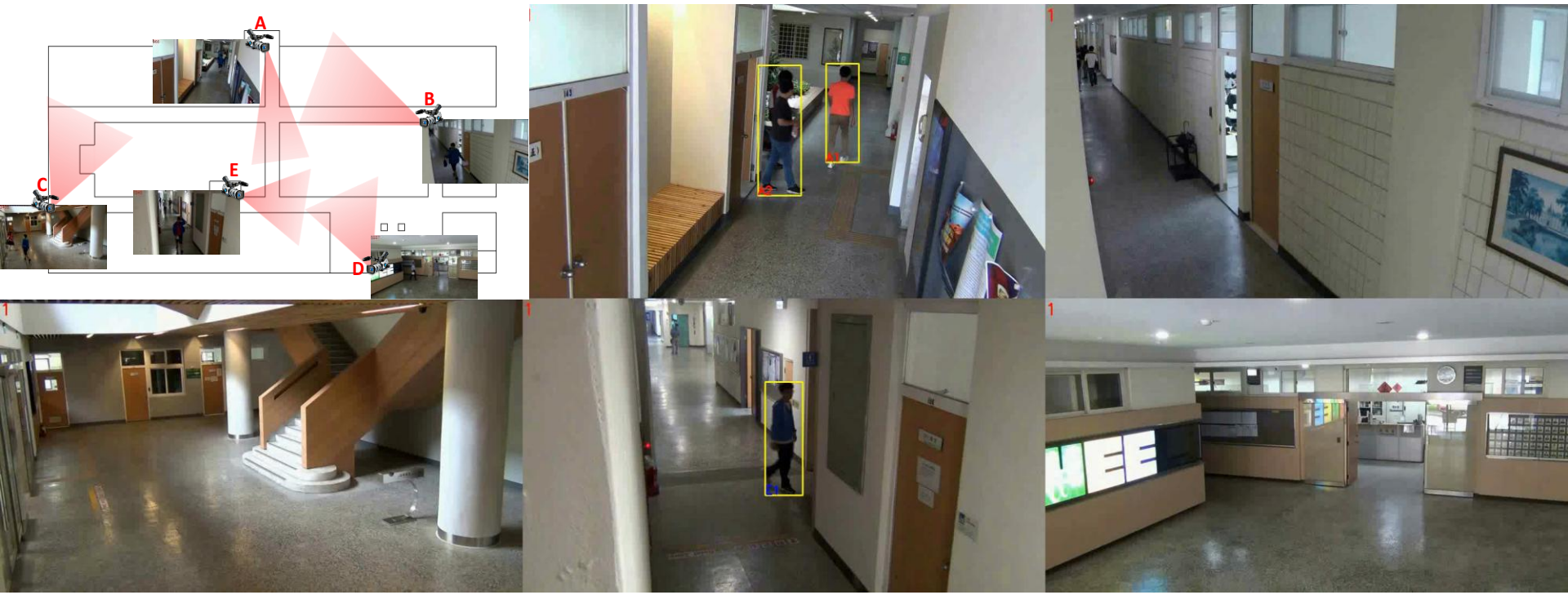
- Smart cars: ADAS



[Intel Mobileye]

Wide Applications of Computer Vision

- Surveillance system



Ref: Chih-Wei Wu, Meng-Ting Zhong, Yu Tsao, Shao-Wen Yang, Yen-Kuang Chen, and Shao-Yi Chien, "Track-clustering Error Evaluation for Track-based Multi-camera Tracking System Employing Human Re-identification," *CVPR 2016 Workshop*.

Wide Applications of Computer Vision

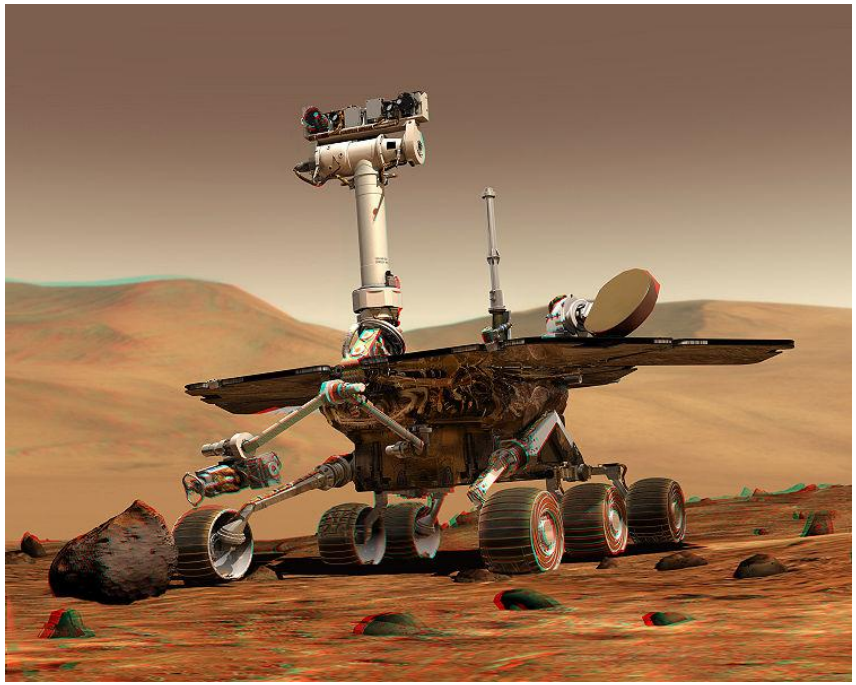
- Vision-based interaction



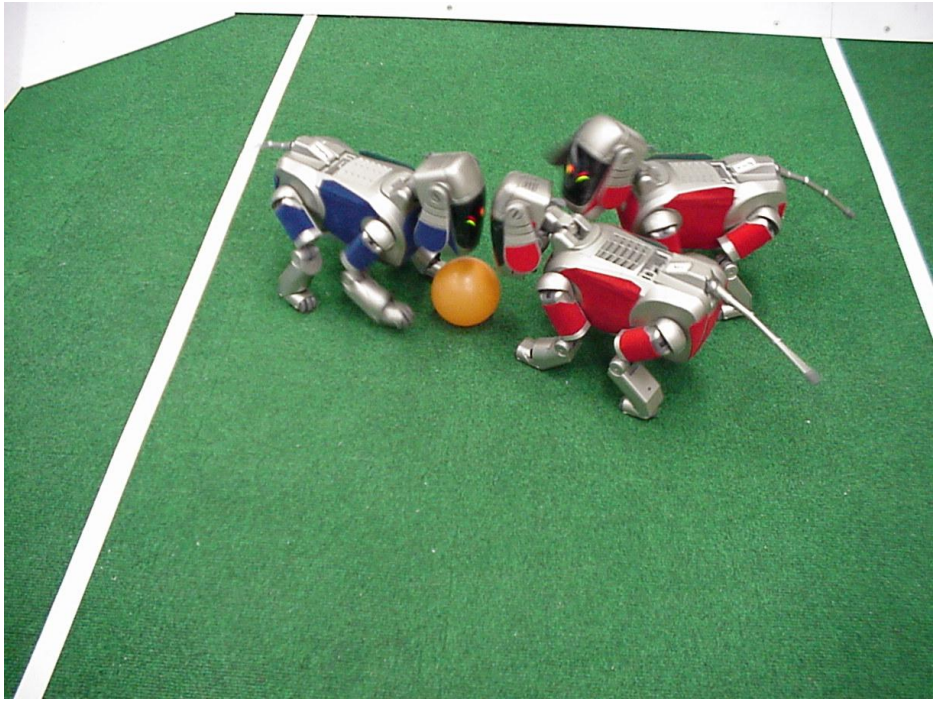
[Microsoft Xbox]

Wide Applications of Computer Vision

- Robotics



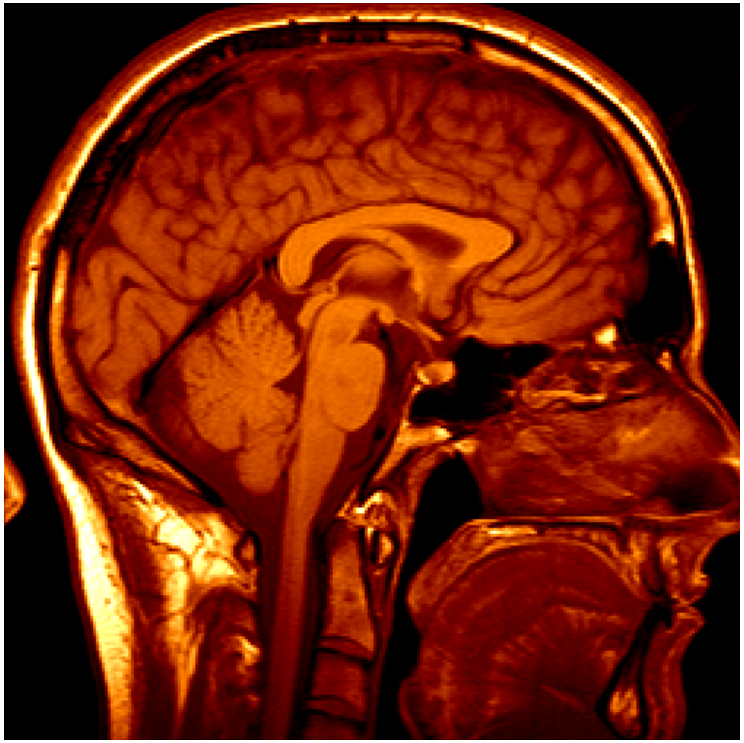
NASA's Mars Spirit Rover
http://en.wikipedia.org/wiki/Spirit_rover



http://www.robocup.org/

Wide Applications of Computer Vision

- Medical image



3D imaging
MRI, CT



Image guided surgery
Grimson et al., MIT

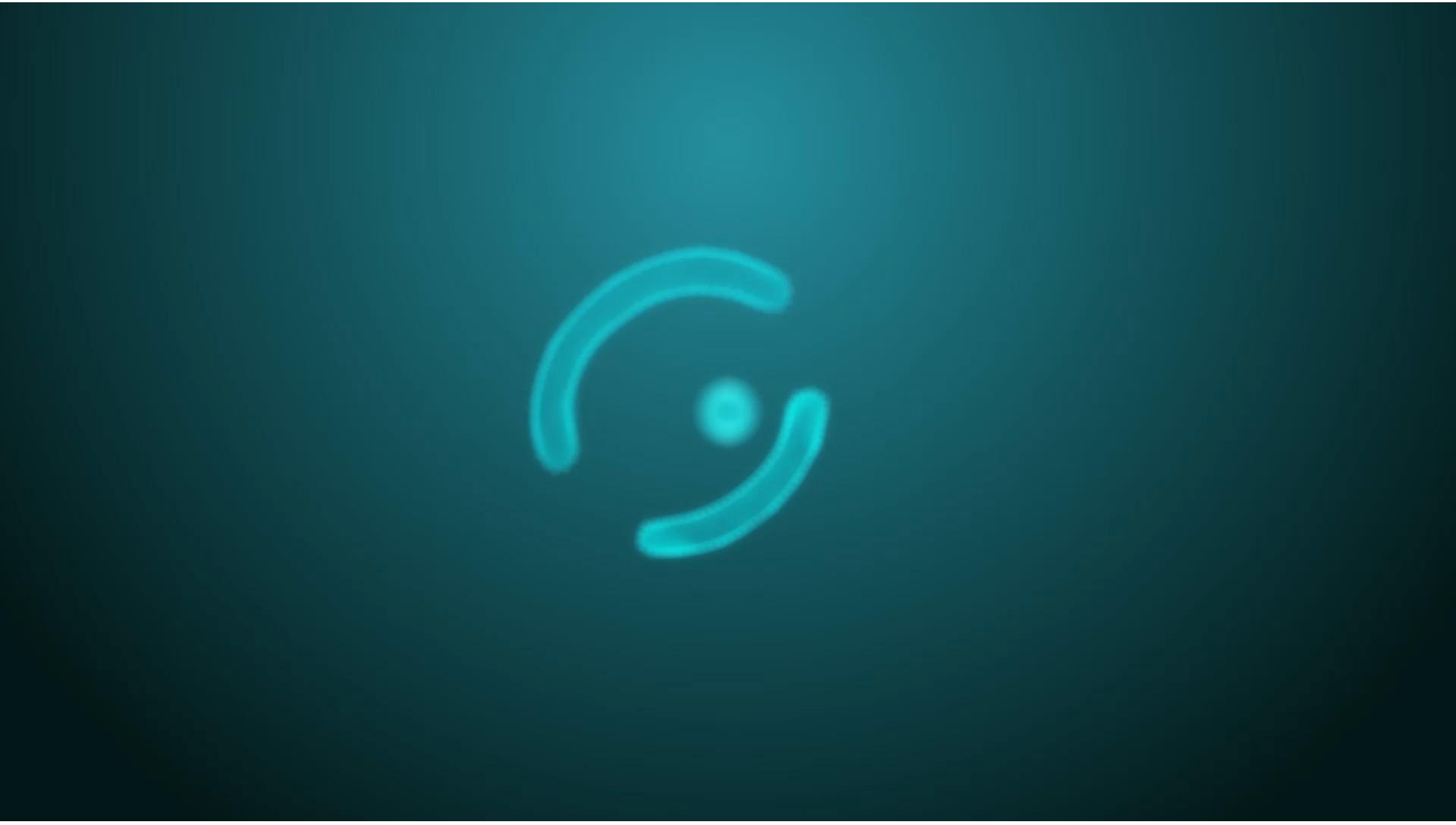
Wide Applications of Computer Vision

[Ganzin]



Wide Applications of Computer Vision

[Ganzin]





打造鋼鐵人、柯南眼鏡不是夢！台大師生看好「眼球追蹤」技術齊心創業

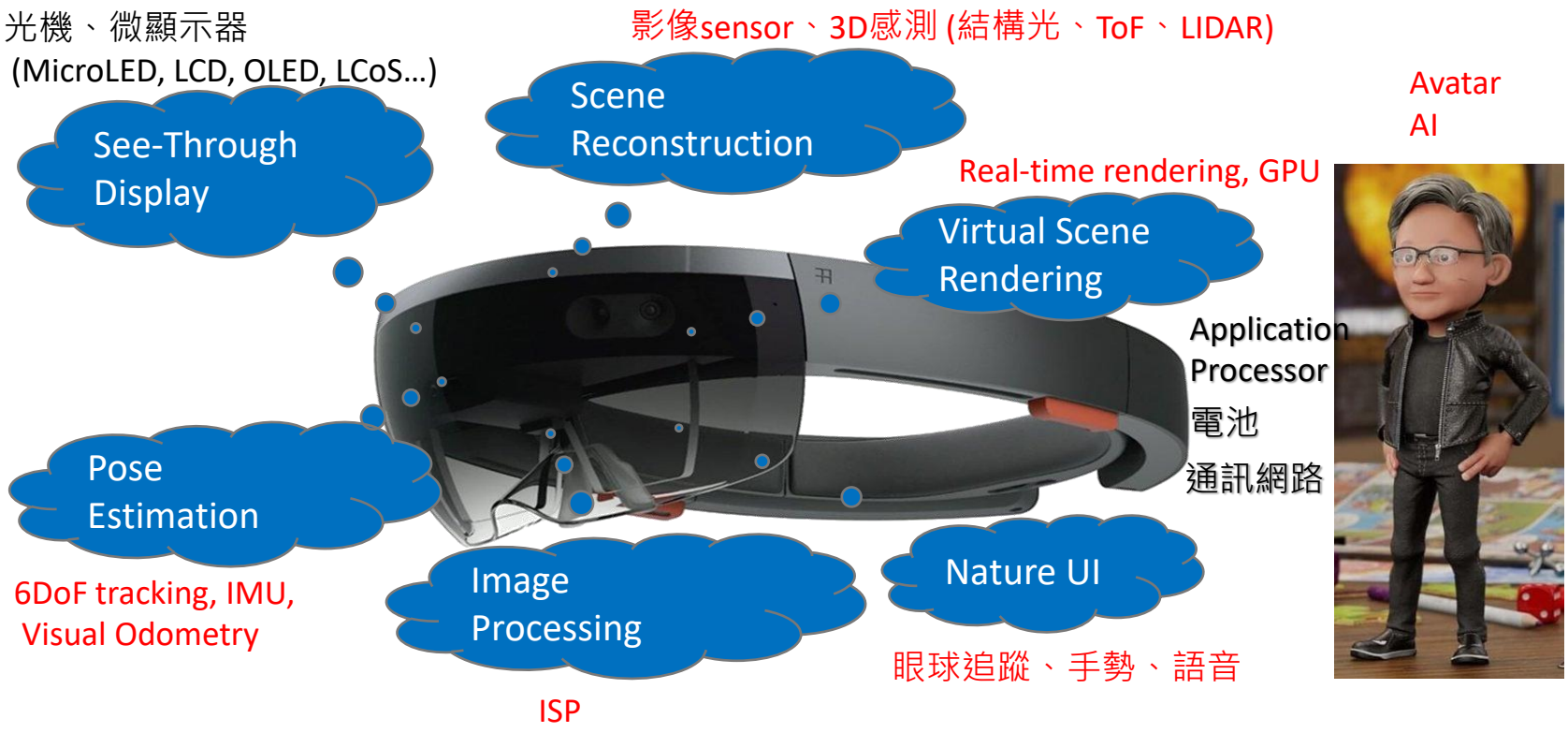
---數位時代創業小聚2020/07/20

ganzin AURORA IIS
AR Glasses Reference Design



Wide Applications of Computer Vision

- AR/VR AI glasses



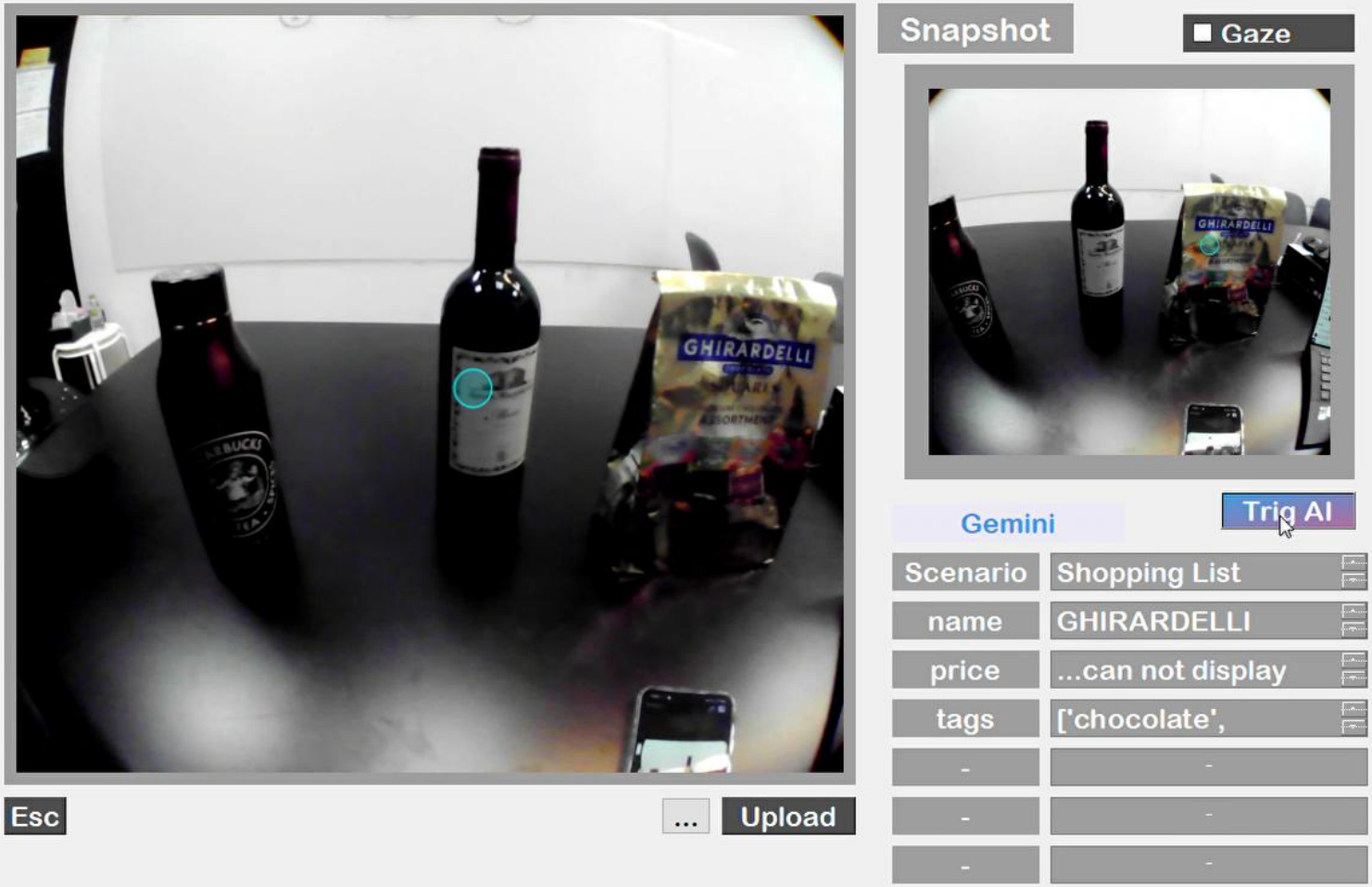
Wide Applications of Computer Vision

[Apple]



Wide Applications of Computer Vision

[Ganzin]



Wide Applications of Computer Vision



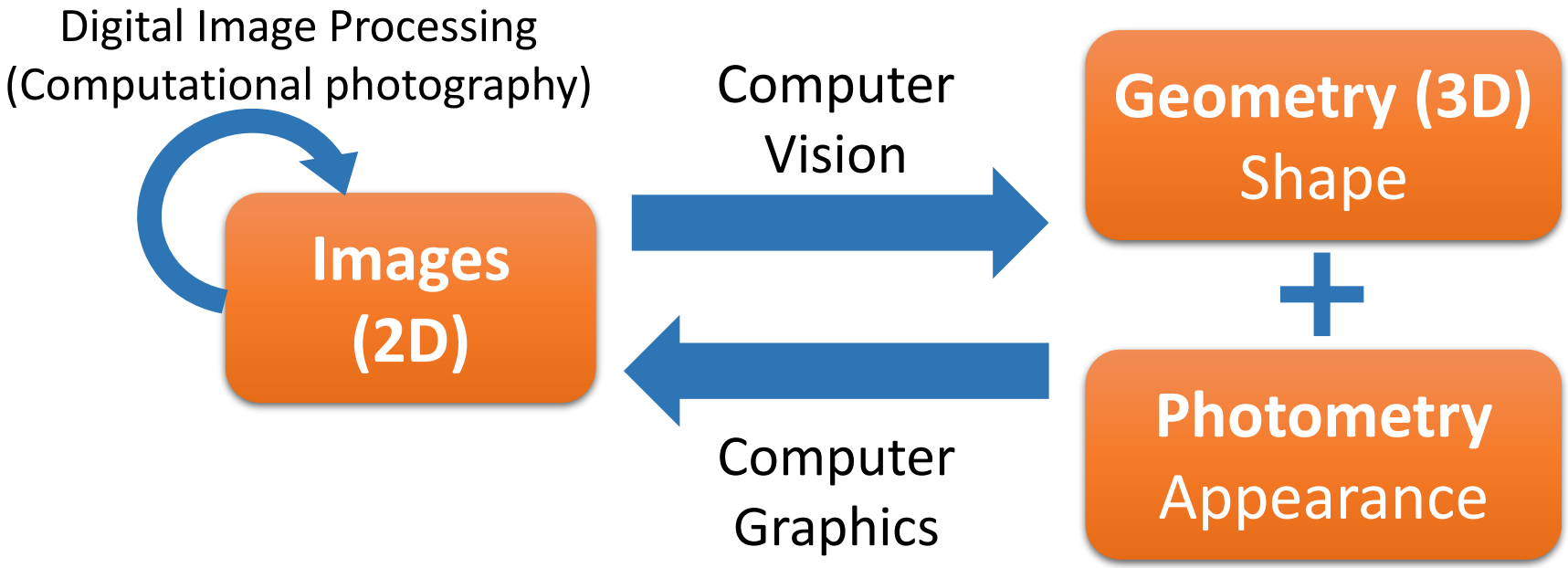
[Sora]

Prompt: A Chinese Lunar New Year celebration video with Chinese Dragon.

Important Near-Future Applications

- AR/VR, Metaverse, spatial computing, AI smart-glasses
- AI agent
- Autonomous vehicles
- Content generation
- Robot
- IoT: AIoT (AI+IoT), IoVT (Internet-of-Video-Things)
- Medical imaging
- Large-scale video analysis
- Computational photography/image synthesis
- Industrial automation
- …

Related Fields



- The boundaries between digital image processing/computer vision/computer graphics become vague nowadays

About this Course…

- Provide a comprehensive introduction to the field of computer vision (CV)
  - From classical methods to deep learning based methods
  - From recognition to geometry
  - No experiences in CV and image processing are required
- The two courses, Computer Vision and Deep Learning for Computer Vision, can give you a complete view of modern CV techniques
- Grading
  - Four homeworks: 60%
  - Class/talk participation: 10%
  - Group final project: 30%

Course Website

- Course website
  - https://cool.ntu.edu.tw/courses/61640
  - 採線上線下混合教學方式，請密切注意NTU Cool訊息
- TA
  - 于富昇 fushengyu@media.ee.ntu.edu.tw

(Tentative) Schedule: May be Modified…

Week	Date	Topic
1	2/27	228和平紀念日
2	3/6	Introduction to human vision systems
3	3/13	Camera basic, image formation and basic Image processing/Feature detection and matching
4	3/20	Machine learning basics
5	3/27	Deep learning basics
6	4/3	春假
7	4/10	Segmentation
8	4/17	Projective Geometry
9	4/24	Estimation of Transformations
10	5/1	勞動節
11	5/8	Single Camera Geometry/Camera calibration
12	5/15	Two-View Geometry
13	5/22	Dense motion estimation/stereo
14	5/29	Optical flow + object tracking
15	6/5	3D reconstruction/depth sensing/Structure from motion
16	6/12	Final project presentation

原則上每兩週會有一次實體上課，暫訂為上表標紅字之日期

Homeworks

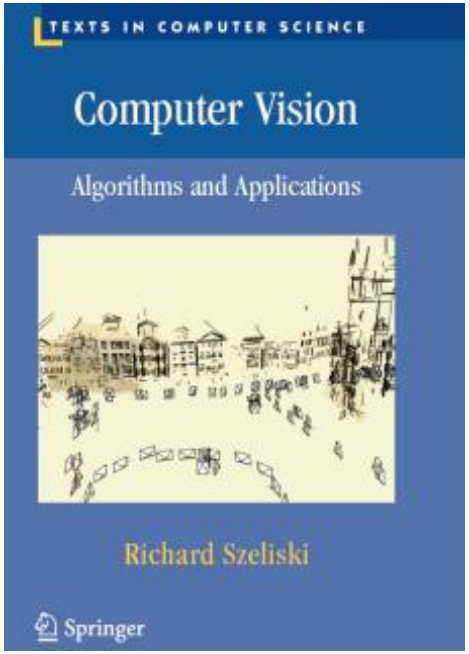
- Four assignments (tentative):
  - HW1: Image filters
  - HW2: Detection or recognition
  - HW3: Pose estimation
  - HW4: Stereo matching (HW3/4 may be merged)
- Official language is Python
- Lab0: Python and basic image processing
  - Will announce later in the next week

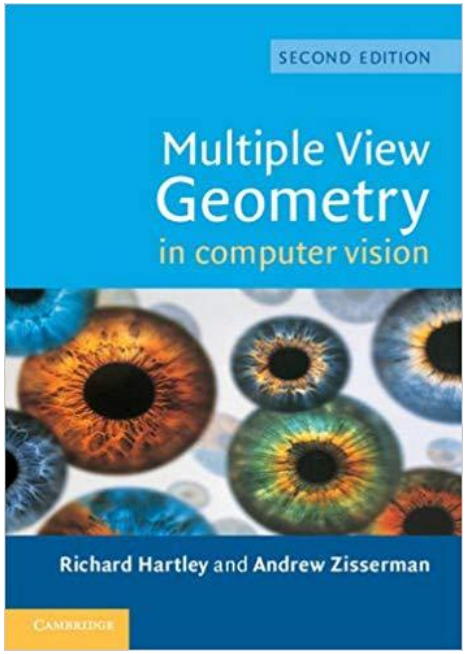
Final Project

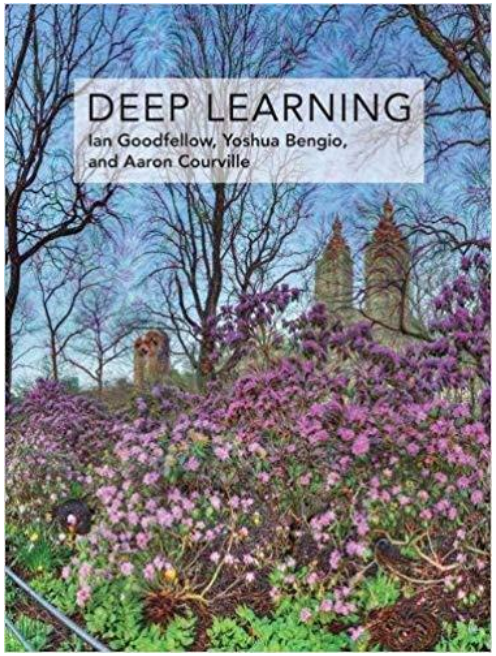
- Will have one or two problems/challenges
- Each team should have 3—4 members
- Project may be supported by industry with awards
- Evaluated by professor, TAs, guest judges from industry, and you (peer review)!
- The problems/challenges will be announced around the week of mid exam

Reference Materials

- Reference books







http://szeliski.org/Book/

- And papers in CVPR, ICCV, ECCV, BMVC, WACV, ACCV, ….

加簽規則



- 請慎重考慮……
- 以教室容量為限，可加簽約40位同學
- 篩選順序
  - 電資學院及重點科技學院(含輔系) > 工學院 > 理學院 > 其他
  - 博班 > 碩二 > 碩一 = 大四 > 大三 > 大二 > 大一
- 想加簽的同學，請於第二節上課之前填寫好下列表單：
  - https://forms.gle/Nn8fGqMN83RXKnQY9
  - Email請務必正確輸入
  - 有選上的同學將在週末寄送授權碼，請馬上加選